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Quantitative Investment Strategies

Beyond Carry: Building a More Resilient FX Return Engine

FX systematic investing has delivered attractive returns over recent years, driven by the re-emergence of carry opportunities and technical return patterns. Extending signal breadth using market insights improves performance further.

QuantWise 

Systematic FX investing has re-emerged as an attractive source of diversified returns as rate and currency dispersion has returned. In FX Carry, flat performance has given way to a more supportive backdrop, especially in developed markets. Over the past year, FX Carry returned 5.1% in Developed Markets (DM) and 8.1% in Emerging Markets (EM). Looking more broadly, realised skewness was even stronger than historically at 10.7% in DM and 25.7% in EM.

An FX Multi-Factor approach provides attractive and robust performance across regimes. By combining technical, fundamental, and sentiment signals, the strategy can offset weakness in one factor with strength in another. Overall, the strategy achieves a Sharpe ratio of 1.87 with a maximum drawdown of -6.8% at 5% annualized volatility. During the Global Financial Crisis, the strategy returned 29.6% and continued to be defensive with a 4.9% return during the distress period in 1H 2022.

The portfolio's performance is not tied to one macro state, but varies in economically intuitive ways. The combined portfolio produced Sharpe ratios of 1.84 when the dollar was rising and 1.70 when it was falling, implying that dollar direction changes factor leadership but returns are robust. The strongest performance came during US growth contractions.

FX Multi-Factor behaves like a genuine diversifier and adds more value than generic currency exposure. Its average absolute correlation with equity, rates, commodity, and volatility strategies is only 0.03, and correlation versus outright equity just 0.04. In an equal-risk multi-strategy portfolio, adding our flagship FX Multi-Factor lifted the Sharpe from 2.80 to 3.29, while adding a classic FX baseline reduced it to 2.70. Adding FX Multi-Factor to a 60/40 portfolio improves performance and crisis resilience.

FX systematic signals provide important inputs for fundamental investors. Systematic screens can support discretionary PMs through robustness checks and guards against bias. The FX Multi-Factor Strategy hold a modest long USD position, though we expect a more dovish Fed to push USD down. Largest current longs of the strategy are IDR, CAD, and TWD while CNH, HKD and ILS are sizable shorts.

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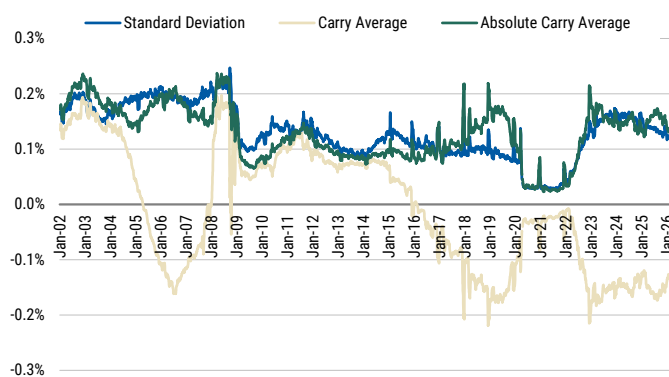
Introduction

We would like to thank Wenlu Sun in Morgan Stanley's IED QIS Structuring group for his contributions to the analyses. Please note that he is not part of Morgan Stanley's Research department.

FX systematic investing is experiencing a renaissance. Amongst the broader set of FX signals, FX Carry¹ has been one of the more profitable investment strategies over the recent years, returning to the strength it showed in the early noughties. More broadly, we find strong returns in a portfolio with an extended set of signals across technical, fundamental, and sentiment metrics. Those returns are resilient over the long run through different market regimes. Systematic FX signals can provide a valuable starting point for fundamental investors in their investment decisions.

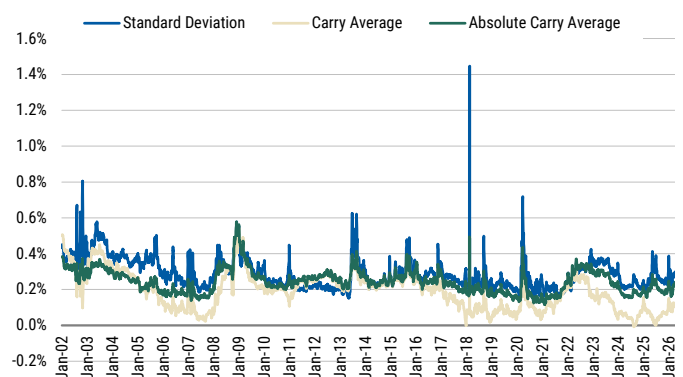
FX Carry is one of the most commonly used systematic investment strategies. In its base form, the strategy borrows money in lower-yielding currencies and invests it in higher-yielding currencies. Implementations can be done using actual physical bonds or currency forwards. The concept has been around for decades, with academic publications studying the performance of the strategy extensively. The uncovered interest rate parity proposes that currencies lose value to the same extent as investors gain through the interest rate differential. This relies on the assumption that monetary policy rates and inflation are correlated. However, the positive long-term performance of FX Carry strategies gives clear evidence that the uncovered interest rate parity does not hold. Academics have studied this phenomenon extensively and attribute the positive returns of FX Carry to risk premia linked to consumption (Lustig, Roussanov, and Verdelhan, 2011), global FX volatility (Menkhoff, Sarno, Schmeling, and Schrimpf, 2012), and crash risk (Chernov, Graveline, and Zviadadze, 2018).

Exhibit 1: Carry opportunities in developed FX markets (DM) over time



Source: Morgan Stanley Research. Note: Averages and standard deviation are calculated in the cross-section of currencies.

Exhibit 2: Carry opportunities in emerging FX markets (EM) over time



Source: Morgan Stanley Research. Note: Averages and standard deviation are calculated in the cross-section of currencies.

FX DM Carry saw a period of flat performance from 2008 to 2020. Many investors turned their backs to FX Carry investing in this period. However, performance over recent years has been persistent and strong. That is true particularly for developed markets. We attribute this strong performance to the increased dispersion in rates across the globe.

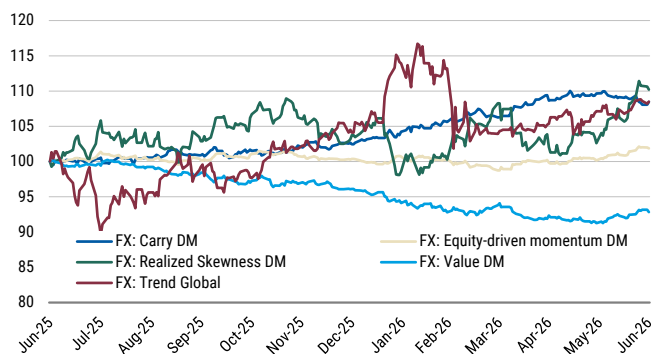
[Exhibit 1](#) and [Exhibit 2](#) show that levels of carry as well as its dispersion have increased

1. In this report, we define carry as the forward-implied carry of a currency versus the USD.

versus its lows, leading to an overall larger amount of carry that can be earned in financial markets. Particularly in the DM space this is the case. Additionally, we saw a strong depreciation of the Japanese yen over the past years, which provided additional tailwinds to investors in FX Carry. We expect this dynamic in FX Carry to continue but less strongly.

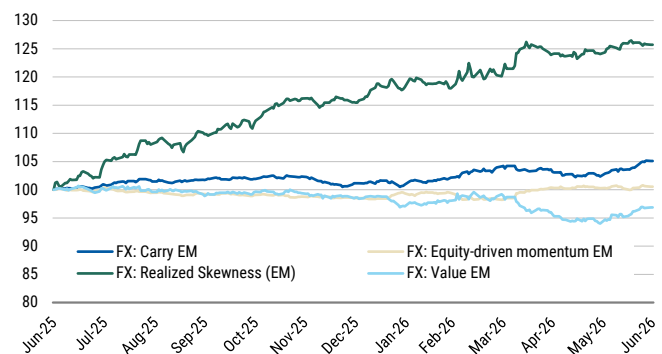
Strength in systematic FX investing is something we have seen more broadly over recent years, particularly the past 12 months. FX Carry up in EM and DM by 8.1% and 5.1%, respectively. Realized skewness, which carries elements of value in it by benefitting from mean reverting markets, had an outstanding performance of 10.7% for DM and 25.7% for EM, which is substantially stronger than we have seen in the past (compare our publication [Quantifying Asymmetry – Realized Skewness as a Core Portfolio Strategy](#) for more detail). Spill-overs from equity markets as we have studied in our publication of equity-driven momentum ([FX Equity-Driven Momentum](#)) are also positive with 2.0% and 0.6% for DM and EM, respectively. The one type of strategy that struggled was actual value. This headwind to FX Value might be turning, though, as the market becomes more focused on valuations. Compare [Exhibit 3](#) and [Exhibit 4](#) for details.

Exhibit 3: Overall attractive performance in DM...



Source: Morgan Stanley Research

Exhibit 4: ... is mirrored in the EM space



Source: Morgan Stanley Research

It does not come as a surprise that the aggregation of a broad set of FX systematic strategies leads to an even stronger risk-adjusted return. Our FX Multi-Factor strategy has achieved Sharpe ratios of ~1.9 since 2005. This performance is achieved by combining the usual carry, fundamental, and technical factors into a balanced multi-factor strategy. Thus, markets are pricing FX signals, leading to strong performance of Quantitative Investment Strategies (QIS) in the FX space.

These results paint an overall positive picture of investing in systematic FX strategies. Forward-looking performance remains strong in our view as carry is comparably attractive, trends continue, and valuations likely to normalize for some currencies (compare, for example, our yen forecasts).

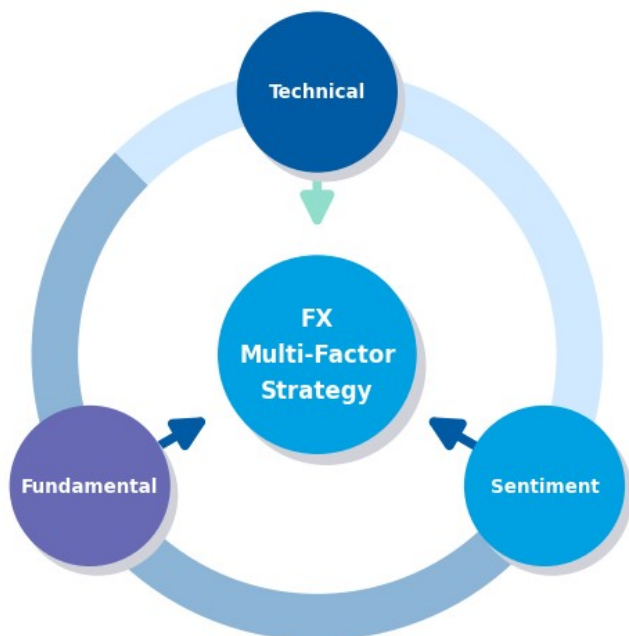
While the focus of this report is on systematic investing, we also highlight that fundamental investors can use the discussed signals to their advantage. Having a dashboard with systematic indicators can help them with initial screens for investment

decisions, guard against confirmation biases, or size trades. We advocate from a fundamental angle for currency carry investing, favouring currently an equally weighted basket of USD, EUR, and JPY to fund an index-weighted basket of EM currencies.

In the remainder of this report, we discuss the return characteristics of our FX Multi-Factor strategy in section [FX Multi-Factor Investing Robust Across Market Regimes](#) . We then turn towards the benefits of integrating FX Multi-Factor investing into portfolios of QIS or traditional 60:40 portfolios in [Cross-Asset Diversification](#) . Our angle then shifts to [The Fundamental View](#) where we discuss how fundamental investors can use this information. We close the analysis with the [Conclusion](#) .

FX Multi-Factor Investing Robust Across Market Regimes

Exhibit 5: FX Multi-Factor



Source: Morgan Stanley Research

We generally prefer a multi strategy approach as it delivers more diversification. Particularly, value strategies tend to complement more pro-cyclical strategies such as FX Carry. Thus, we turn in this section to our multi-factor FX strategy, which incorporates a broader set of signals that can be categorized into (1) technical, (2) fundamental, and (3) sentiment signals (compare [Exhibit 5](#)). This strategy has had a very strong long-term return overall and continuing through recent years. We focus in particular on the performance of the FX Multi-Factor Strategy across different market regimes.

At a high level, we score every currency on each signal group, turn those scores into long and short positions, and blend them into one systematically rebalanced portfolio scaled to a consistent risk target. Because the signals capture distinct, loosely-related drivers of currency returns, combining them smooths performance and avoids relying on any single factor. We then examine how this diversified strategy behaves across different market regimes.

Performance overview

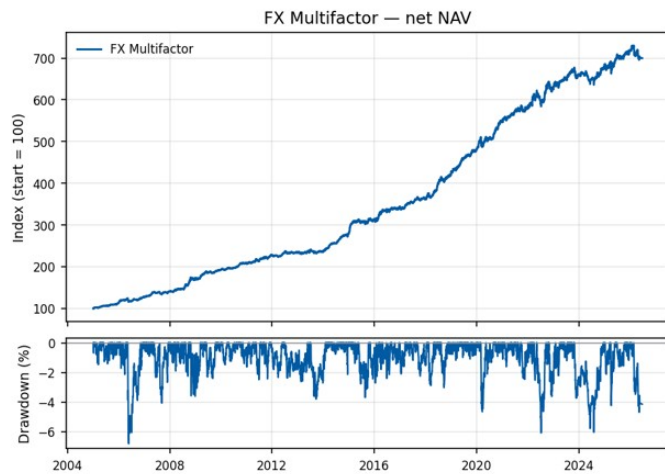
The combined FX Multi-Factor portfolio delivers a Sharpe ratio of 1.87, with a CAGR of 9.3% and a maximum drawdown of -6.8%. The result reflects the central design choice of the strategy: diversification across several low-correlated factors that draw on distinct return drivers, so the portfolio has more than one way to earn returns and to weather drawdowns. The returns are persistent over time without any prolonged down periods as can be seen in [Exhibit 6](#).

Resilience in market distress

FX Multi-Factor is actually a strong Diversifier strategy in market distress. We use the same Drawdown Coverage (DDC)² metrics used in our previous report on Risk Mitigation Portfolios ([A New Paradigm for Risk Mitigation in Pension Plan Portfolio Construction](#), March 20, 2026) to evaluate the behavior of our FX Multi-Factor strategy in market distress periods. We find that the DDC comes out at 1% for daily equity market drawdowns and 4% for quarterly equity market drawdowns (compare [Exhibit 7](#)). Thus, the strategy has a marginally defensive tilt to it but is overall rather immune to equity market drawdowns. This characteristic indicates that the FX Multi-Factor strategy might be a good diversifier in traditional portfolios.

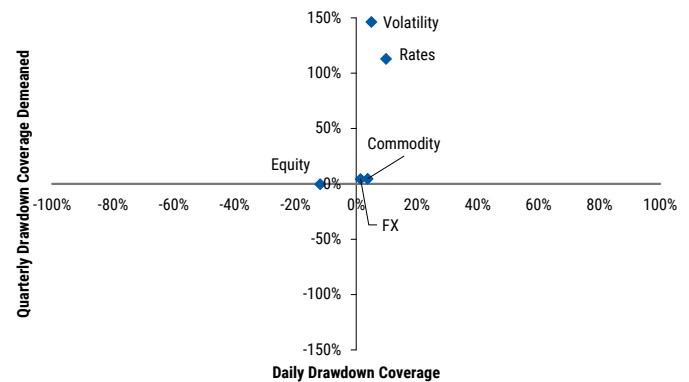
2. DDC is defined as minus one times the average returns of an investment strategy divided by the average returns of a benchmark. Observations are conditioned to periods where the benchmark is down one standard deviation or more. We use the MSCI World as benchmark. The metric is calculated for daily/quarterly returns.

Exhibit 6: The FX Multi-Factor portfolio compounds steadily across the sample with no single dominant regime



Source: Morgan Stanley Research

Exhibit 7: Performance of Multi-Factor Quantitative Investment Strategies by Asset Class in Distress Periods



Source: Morgan Stanley Research

We first study returns of the combined portfolio over notable historical episodes (the Global Financial Crisis, the 2013 taper, the COVID shock, the 2022 hiking cycle, the 2025 tariff shock, and the 2026 Middle East conflict) to see how it behaved through known stress windows. Returns are cumulative within each window while volatility, Sharpe ratio, and drawdowns are annualised on the same 5% volatility basis as the rest of the publication.

Exhibit 8: FX Multi-Factor Portfolio Performance Over Notable Historical Episodes

Episode	Return	Ann. vol	Sharpe ratio	Max dd	Interpretation
GFC 2008	29.6%	6.9%	2.34	-4.1%	Strong crisis performance
VIX Crash 2018	1.6%	4.3%	11.78	-0.2%	Contained through the volatility spike
Q4 2018	3.1%	4.6%	3.11	-0.9%	Resilient in the risk-off selloff
COVID-19	-2.4%	10.3%	-2.23	-4.7%	Positive but volatile
January-June 2022	4.9%	6.5%	1.69	-2.7%	Benefited from macro dispersion
Trump Liberation Day	-2.4%	20.3%	-3.81	-3.4%	Drawdown then swift recovery
Iran Conflict 2026	-1.9%	7.1%	-3.11	-2.1%	Contained drawdown

Source: Morgan Stanley Research

The portfolio stays resilient across distress periods — drawdowns in individual stress windows are moderate, suggesting that historical stress has not been tied to one recurring weakness in the framework. The pattern across windows is informative: the portfolio held up well through the 2008 Global Financial Crisis and the 2022 rate hiking cycle, and was positive but choppy through the 2020 COVID shock.

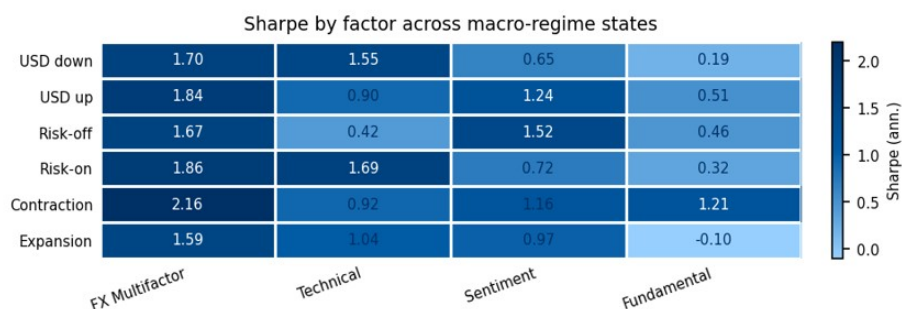
The two most recent shocks make the same point about why the portfolio holds up — it does not lean on a single defensive factor, but on dispersion across factors. The 2026 Middle East conflict is the most recent geopolitical stress in the sample and resulted in marginally negative returns for the strategy.

Macro regimes

We condition performance on three macro regimes:

- **USD direction (USD up / USD down):** the trade-weighted dollar (DXY) relative to its trailing one-year trend — USD up when DXY is above its 1-year moving average, USD down otherwise.
- **Risk sentiment (Risk-on / Risk-off):** a global equity index relative to its trailing one-year trend — Risk-on above the 1-year moving average, Risk-off below.
- **US growth (Expansion / Contraction):** US manufacturing PMI relative to the 50 threshold — Expansion at or above 50, Contraction below.

Exhibit 9: Sharpe of the combined portfolio and the four factors across macro-regime states



Source: Morgan Stanley Research

The performance heatmap in [Exhibit 9](#) is consistent with the economic design of the three factors. Technical tends to do better in benign or USD-down environments, when funding conditions are easy and high-yielders are bid. Sentiment tends to contribute more in risk-off or USD-up environments, when positioning and cross-market risk signals matter most. Fundamental signals are on average positive in Sharpe ratio but weaker than the other categories. However, they are comparably strong in contractions, making them defensive signals.

Exhibit 10: Combined FX Multi-Factor Strategy by regime state, across all three macro regimes

Regime	State	Sharpe ratio	Ann. return	Ann. volatility	Days
USD direction	USD down	1.70	8.1%	4.7%	2777
USD direction	USD up	1.84	9.7%	5.2%	2826
Risk sentiment	Risk-off	1.67	10.7%	6.4%	1356
Risk sentiment	Risk-on	1.86	8.3%	4.4%	4247
US growth (PMI)	Contraction	2.16	12.1%	5.6%	895
US growth (PMI)	Expansion	1.59	7.5%	4.7%	4080

Source: MS Research.

USD direction. The combined portfolio records a Sharpe of 1.84 in its stronger USD state (USD up) versus 1.70 in the weaker one (USD down). USD direction matters for FX factors because the dollar is the common leg of most crosses: it drives funding conditions and the risk appetite that carry and sentiment depend on.

Risk sentiment. Across risk states the portfolio holds up — a Sharpe of 1.86 in Risk-on versus 1.67 in Risk-off. The risk-on / risk-off label is the global equity index versus its 1-year trend, an ex-post classification that would lag in real time.

US growth (PMI). The portfolio earns a Sharpe of 2.16 in Contraction versus 1.59 in Expansion. We use US manufacturing PMI because it is a timely, globally-watched gauge of the manufacturing and trade cycle that drives many of the currencies in the portfolio.

USD exposures

In our portfolio construction, the FX Multi-Factor can take residual USD exposures. We run analyses to evaluate if there is value in those exposures. To test this, we treat the dollar leg as a standalone strategy — each day we hold the portfolio's net dollar exposure as a position in the Bloomberg Dollar Total Return Index (a weighted basket of rolling one-month US dollar forwards) and earn that exposure times the index's return. The position is taken with a one-day execution lag. Unlike every other result in this report, this exhibit is shown gross of costs: it applies no transaction costs and is intended only as an illustration of the information in the dollar tilt, not as a tradable, net-of-costs result. On that gross basis, and over the sample, the dollar leg delivers a Sharpe ratio of 0.36, with a CAGR of 1.8% (compare [Exhibit 11](#) and [Exhibit 12](#)). Thus, there is value in taking more directional exposures and de-monopolizing the USD as numeraire.

Exhibit 11: Cumulative return of holding the FX Multi-Factor portfolio's net US dollar exposure



Source: Morgan Stanley Research

Exhibit 12: Performance of the dollar-leg strategy

Strategy	Ann. return	Ann. volatility	Sharpe ratio	Max drawdown
Dollar leg (net USD exposure)	1.8%	5.0%	0.36	-18.0%

Source: Morgan Stanley Research

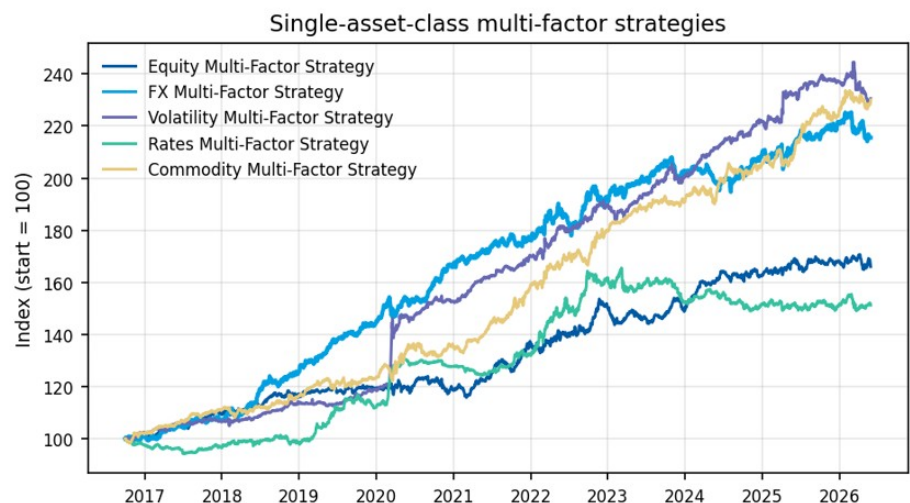
Cross-Asset Diversification

The preceding sections argued that the FX Multi-Factor portfolio earns its return from currency-specific risk premia that are largely distinct from the broad market. If that is correct, the portfolio should behave very differently from the other systematic strategies an investor might already own. We test that here by measuring how the portfolio co-moves with a set of other cross-asset strategies, both on average over the full sample and as that relationship evolves through time.

Single-asset-class multi-factor strategies

We compare five single-asset-class multi-factor strategies — this report's FX Multi-Factor Strategy, an Equity Multi-Factor Strategy, a Volatility Multi-Factor Strategy, a Rates Multi-Factor Strategy, and a Commodity Multi-Factor Strategy. Each applies a disciplined multi-factor approach within its own asset class. The chart in [Exhibit 13](#) and table in [Exhibit 14](#) below show all five over the common window in which they are all available (September 2016 to May 2026), each scaled to the same volatility so their risk-adjusted returns are directly comparable.

Exhibit 13: Cumulative return of the five single-asset-class multi-factor strategies (FX / Equity / Volatility / Rates / Commodity), each scaled to a 5% annualized volatility



Source: Morgan Stanley Research

Exhibit 14: Standalone performance of the five single-asset-class multi-factor strategies (net of costs)

Strategy	Ann. return	Ann. volatility	Sharpe ratio	Max drawdown
Equity Multi-Factor Strategy	5.3%	5.0%	1.06	-6.6%
FX Multi-Factor Strategy	8.2%	5.0%	1.63	-6.4%
Volatility Multi-Factor Strategy	8.9%	5.0%	1.78	-6.6%
Rates Multi-Factor Strategy	4.3%	5.0%	0.86	-10.0%
Commodity Multi-Factor Strategy	8.9%	5.0%	1.77	-5.0%

Source: Morgan Stanley Research

All five strategies deliver positive risk-adjusted returns over the common window, which is consistent with the idea that a disciplined multi-factor approach travels across asset classes rather than being specific to any one market. This report's FX Multi-Factor Strategy earns a Sharpe ratio of 1.63 over this window. The strongest of the five is the Volatility Multi-Factor Strategy at a Sharpe ratio of 1.78. More important for what follows is that these are genuinely different return streams: the next section shows they are only loosely correlated with one another, which is what makes combining them worthwhile.

Correlation with other strategies

We compare the correlations of the multi-factor strategies (see [Exhibit 15](#)) using daily returns over the common window in which all series are available (September 2016 to May 2026).

Exhibit 15: Pairwise correlation of daily returns — the FX Multi-Factor portfolio and four other cross-asset systematic strategies

	FX Multi-Factor Strategy	Equity Multi-Factor Strategy	Rates Multi-Factor Strategy	Commodity Multi-Factor Strategy	Volatility Multi-Factor Strategy
FX Multi-Factor Strategy	1.00	0.04	0.03	-0.03	-0.03
Equity Multi-Factor Strategy	—	1.00	0.02	-0.02	0.07
Rates Multi-Factor Strategy	—	—	1.00	0.01	0.06
Commodity Multi-Factor Strategy	—	—	—	1.00	-0.05
Volatility Multi-Factor Strategy	—	—	—	—	1.00

Source: Morgan Stanley Research

The FX Multi-Factor portfolio shows low correlation with every other strategy in the table. Its average absolute correlation with the other strategies is 0.03. Its strongest single relationship, with the equity strategy, is modest at 0.04. We read this as consistent with the economic thesis of the report: the portfolio's returns come from unique FX performance drivers rather than from broad equity, duration, or commodity factors.

A single full-sample number can hide periods where a relationship tightens — for example during a shared market stress. Thus, we track the correlation through time: using weekly (five-day) overlapping returns, we compute the correlation of the FX Multi-Factor portfolio with the other strategies over a trailing two-year window. The chart below plots those rolling correlations from September 2018 onward (the first two years are used to seed the initial window).

Exhibit 16: Two-year rolling correlation of the FX Multi-Factor portfolio with each of the four other multi-factor strategies, computed from weekly (five-day) overlapping returns



Source: Morgan Stanley Research

The rolling correlations stay in a narrow band around zero throughout the sample, drifting modestly positive or negative for individual strategies at times but without any sustained move to a high correlation: the largest reading in absolute terms over the whole history is only about 0.45. We read this as evidence that the strategy's independence is a persistent feature rather than an artefact of one calm period — its correlation with these strategies does not spike. This evidence is consistent with [Exhibit 7](#) where we find that FX Multi-Factor is a great diversifier in times of equity market distress. The same is true for Equity and Commodity Multi-Factor strategies. Interestingly, we find in [Exhibit 7](#) that Rates and Volatility Multi-Factor strategies are rather defensive in longer equity market stress periods (i.e., quarterly DDC is significant and positive).

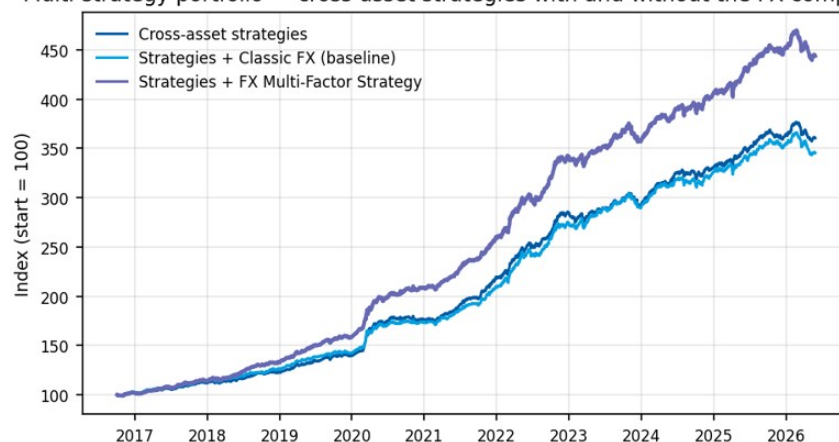
The correlation evidence above is about co-movement and risks. The more practical question is what the strategy does for an investor who already runs the other cross-asset strategies in terms of performance. We answer it by combining those strategies into a single multi-strategy portfolio and check if the inclusion of FX Multi-Factor strategies increases risk-adjusted returns. Also, we compare our main FX Multi-Factor strategy to a more simplified, classic version that just uses baseline signals.

We build three equal-risk portfolios over the common window in which all the series are available (Oct 2016 to May 2026), each scaled to the same 5% volatility so the strategies carry equal risk weight:

- Cross-asset strategies — the four other asset-class strategies (equity, rates, commodity, and volatility) blended together, with no currency component.
- Strategies + Classic FX (baseline) — the same four strategies plus a currency component set to the classic baseline (the classic factor set contains a simplified version of our FX Multi-Factor strategy), each of the five components carrying equal risk.
- Strategies + FX Multi-Factor Strategy — the same four strategies plus the full FX Multi-Factor Strategy as the currency component, again five equal-risk components.

Exhibit 17: Equal-risk multi-strategy portfolios (rebased to 100): the four other cross-asset strategies on their own, those strategies plus the classic FX baseline, and those strategies plus the FX Multi-Factor Strategy

Multi-strategy portfolio — cross-asset strategies with and without the FX component



Source: Morgan Stanley Research

Exhibit 18: Standalone performance of the three equal-risk multi-strategy portfolios (net of costs)

Portfolio	Ann. return	Ann. volatility	Sharpe ratio	Max drawdown
Cross-asset strategies	14.0%	5.0%	2.80	-5.2%
Strategies + Classic FX (baseline)	13.5%	5.0%	2.70	-6.3%
Strategies + FX Multi-Factor Strategy	16.4%	5.0%	3.29	-6.5%

Source: Morgan Stanley Research

The cross-asset strategies portfolio is already strong when excluding FX Multi-Factor, with a Sharpe ratio of 2.80. Adding the classic FX baseline as a fifth component does not improve it — the combined Sharpe ratio is 2.70, essentially unchanged. Thus, a naïve currency strategy brings little incremental risk-adjusted return to a portfolio that is already well diversified. Replacing that classic component with the flagship FX Multi-Factor Strategy changes the picture: the combined Sharpe ratio rises to 3.29, the clearest improvement of the three. The gain reflects the quality of the FX Multi-Factor Strategy's own return stream rather than a change in how the currency component diversifies the portfolio. The takeaway is that it is not currency exposure per se that helps a multi-strategy investor, but this particular currency strategy.

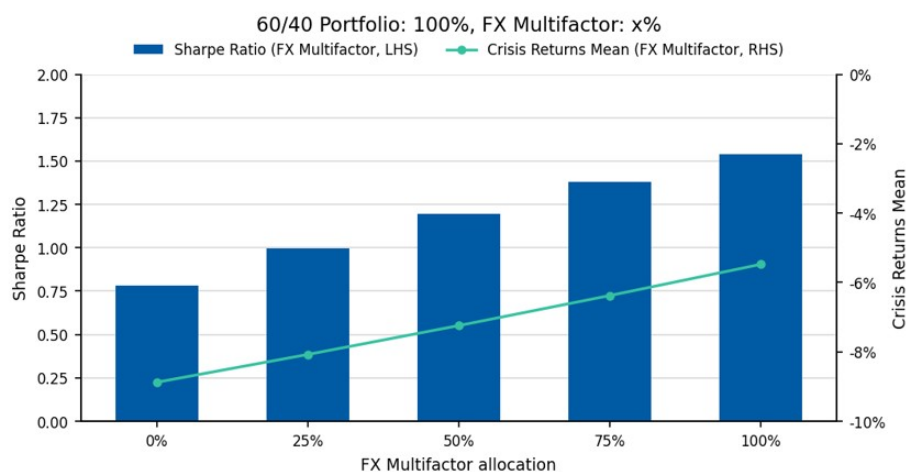
Adding to a classic 60/40 portfolio

Having assessed the benefits of adding the FX Multi-Strategy portfolio to a broad portfolio of multi-strategy alternatives, we are now looking at the impact of the FX Multi-Factor strategy when added to a traditional portfolio. To test that, we build a classic 60/40 multi-asset benchmark and ask whether adding the FX Multi-Factor portfolio to it improves the performance.

We construct the 60/40 from two widely-used total-return market indices: a global developed-market equity index (MSCI World) for the 60% equity portion, and a broad global investment-grade bond index (Bloomberg Global Aggregate) for the 40% bond portion. We rebalance back to the 60/40 mix every month and compound the blended monthly returns into a single portfolio.

Because the FX Multi-Factor Strategy is a long/short overlay that requires little capital, it can be layered on top of the fully-invested 60/40. We keep the 60/40 at its own natural volatility — the benchmark as an investor actually holds it — and size the overlay with increasing allocation from 0% up to 100%. At each step, we analyze its Sharpe ratio over the full common window and average returns during crisis periods (i.e., the cumulative move over each crisis window, averaged across the GFC 2008, VIX Crash 2018, Q4 2018, COVID-19, January-June 2022, Trump Liberation Day, and Iran Conflict 2026 episodes). The chart below plots both together — Sharpe ratio as bars on the left axis, average crisis-period return as a line on the right axis.

Exhibit 19: A 60/40 portfolio at its native volatility with an FX Multi-Factor Strategy overlay sized from 0% to 100%: combined Sharpe ratio (bars, left) and the average cumulative return across crisis periods (line, right)



Source: MS Research.

The Sharpe ratio moves in the investor's favour as the overlay is scaled up, rising from 0.78 with no overlay to 1.54 at the full allocation. The average crisis-period return improves as the overlay is scaled up too, moving from -8.9% with no overlay to -5.5% at the full allocation. The two effects broadly reinforce each other: the overlay adds risk-adjusted return in normal times and, because its returns are largely independent of the 60/40, it tends not to compound the 60/40 portfolio's losses when markets are stressed. The evidence is consistent with the FX Multi-Factor Strategy adding value as a diversifying overlay to a conventional multi-asset allocation.

The Fundamental View

We now turn towards a discussion of how systematic FX investing can be helpful for fundamental investors. We approach this section by focusing on three aspects: (1) how the systematic signal compares with our current views on a) USD and b) CNH (the largest position in the strategy); (2) the general role systematic signals can play in the portfolio of fundamental FX investors, and (3) our current preference for an FX carry trade, which combines systematic signals around carry with fundamental market views.

The Systematic Strategy vs MS views

In terms of current positioning, the FX Multi-Factor strategy has a total exposure of -21.0%, which means that it is long the USD versus the traded currencies in our portfolio. Thus, the strategy is positioned for USD strength. We show current positioning in [Exhibit 20](#). Currently, the strategy is skewed to be short other currencies versus the USD. CNH is the most significant short with -47.2%. On the long side, the largest position is IDR with +30.8%. These top longs and shorts provide a good starting point to build high conviction fundamental long and short positions in an FX portfolio.

Exhibit 20: Current Positioning - top 3 long and short positions

Currency	Weight	Currency	Weight
CNH	-47.2%	TWD	25.5%
HKD	-39.9%	CAD	28.7%
ILS	-30.7%	IDR	30.8%

Source: Morgan Stanley Research

The net long USD position of the strategy is sending a different signal to our current view on USD, where we are neutral but have a bearish bias. In recent weeks, the USD has been strong and so the strategy continues to challenge our thinking and encourages us to look for blind spots, in our view.

Indeed, it is notable that the strategy has held long USD position to varying degrees since mid-March, and the USD is generally higher since then.

A key pillar of our current bearish bias on the USD is the expectation for the Fed Funds rate to be lower than what markets and the broader consensus expect. This more dovish path for US monetary policy is not something that the systematic strategy would necessarily reflect and helps to explain where we differ.

The short position against CNH is notable, as the currency has been among the top performers within Asia so far this year and investors are on the look out for catalysts that could reverse the trend. It's important to point out that weight assigned to each currency is in part a function of its volatility. CNH and HKD receive high weights in part because the volatility is low (clearly so in the case of USD/HKD given it trades within a tight band).

Nonetheless, looking at the trend in the weight can tell us about the strategy's level of conviction in the call, and the size of the short position recommended against CNH has increased sharply in the past few weeks, and so is sending a new signal worth considering.

The continued dependence on the export channel for economic growth likely means that sustained CNY strength and outperformance vs peers poses headwinds for the economy. Recent data show continued increases in the trade surplus, which don't indicate a significant concern for the currency yet, but the systematic Strategy is picking up something that deserves further interrogation.

For a discretionary macro PM looking to incorporate the signals from specific currency weights into an investment decision, it may be necessary to convert the weights to z-scores, or otherwise adjust for the volatility. Over time, we will look to more formally integrate systematic signals into work of the FX Strategy team.

Exhibit 21: Strategy* holds a moderate long USD position today



Source: Morgan Stanley Research *Strategy positions in excess of 100/-100% indicate leverage

How Discretionary Macro PMs Could Use Systematic Strategies

Most of our research in the FX EM Strategy relies on a discretionary approach, with an analysis of business cycles, monetary policy, and risk appetite, mixed in with quantitative tools to help us understand valuation, positioning, and risk premia. In these quantitative tools, considerations around sentiment, technical, and fundamental signals - as we apply in our FX Multi-Factor strategy - play an important role. Thus, we are strong proponents of quantitative inputs in a fundamental decision making process. However, we don't necessarily mix these variables together in a fixed or systematic way, but use our judgement to tie the pieces of the puzzle and place different weights on various components of the story as the cycle evolves. Many of our clients at asset managers and hedges funds will follow a similar approach.

There is, however, value in considering signals from systematic strategies as another input within a broader discretionary approach. Cherry-picking when to incorporate a signal from a systematic strategy is not within the spirit of how systematic strategies are

intended, of course. However, for a discretionary macro portfolio manager, there is no such promise to stick to a systematic model-driven approach in the first place. Using such signal selectively is normal discretionary practice, similar to when incorporating other inputs such as positioning, central bank commentary, or macro data.

When might systematic signals add value to a discretionary process?

- **Guarding against confirmation bias:** Any discretionary process is potentially exposed to confirmation bias. Adding a proven systematic overlay could serve as a tool to guard against this. If a particular currency has been performing well for some time in a discretionary portfolio and an investor is inclined to keep the position, but the systematic strategy suddenly starts sending the opposite signal this can act as a prompt to re-examine the thesis and check for blind spots.
- **Timing & sizing trades:** Validation and support for a particular position by a systematic strategy could help with conviction and thus provide the signal needed to enter or exit a trade, or to adjust the size of it within the overall portfolio.
- **Filtering a market view:** Multi-factor systematic strategies are agnostic to the market regime. However, discretionary portfolio managers will likely take a view on whether markets are trending, in a risk on or off backdrop and thus whether carry trades are likely to be effective. By taking a view, the discretionary macro PM could tune into a particular factor within the multi-factor strategy that he/she expects to be dominant, filtering out those are not likely to apply. If we are in a

range-trading environment, momentum strategies could be filtered out, for example.

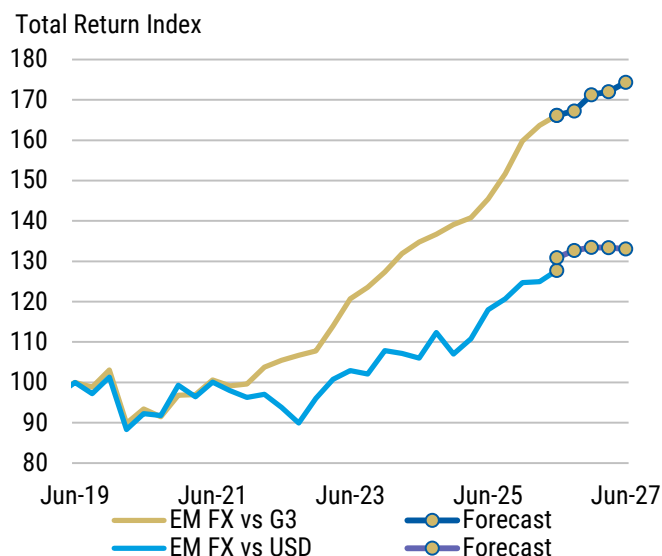
As mentioned previously, within the FX Strategy team we use a range of quantitative signals (valuation signals, risk premia, and positioning indicators, for example) as an input to our process already. Over time, we expect to integrate more factor-based strategies such as those found within this publication into our process.

Finding Your Funder for Carry Trades

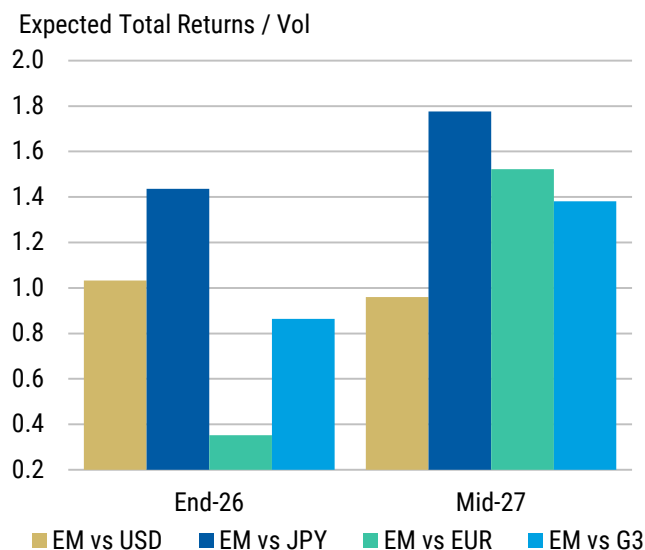
Long EM vs basket: Our forecasts suggest an index-weighted basket of EM currencies, funded by an equally-weighted basket of USD, EUR, and JPY, would generate steady total returns over the next 12m, while a purely USD funded approach would see returns flatline from the end of 2026 ([Exhibit 22](#) & [Exhibit 23](#)).

EUR funding has outperformed: Year-to-date, funding EM via EUR has generated strong total returns with little volatility. We think EUR-funded exposure has been the best approach for investors. Yet, given the challenges that can come with picking the most effective funding currency, a basket approach could similarly offer an attractive return profile for investors, with a G3 funding strategy delivering both attractive absolute returns with a Sharpe ratio only modestly below the narrower EUR-funded approach. ([Exhibit 24](#) & [Exhibit 25](#))

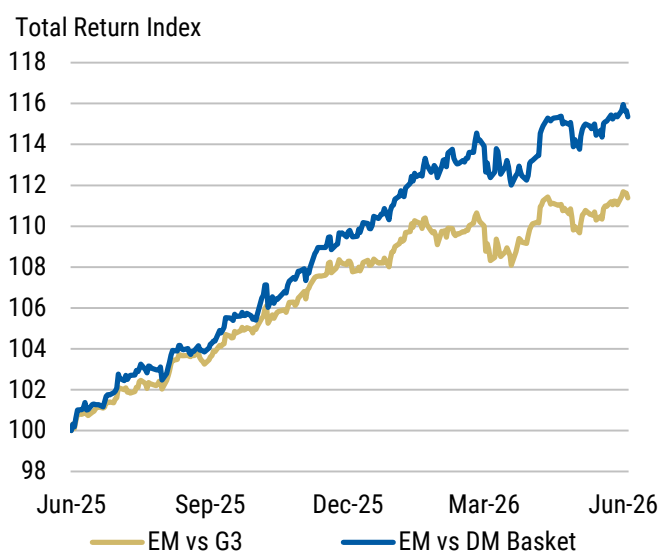
Diversified funding with DM commodity currencies, less attractive: Funding EM exposure via a broader basket of DM of G3 + CHF, AUD and CAD would have delivered a less attractive return profile with AUD helping to reduce volatility relative to USD funding but at the expense of returns.

Exhibit 22: EM gains vs G3 expected

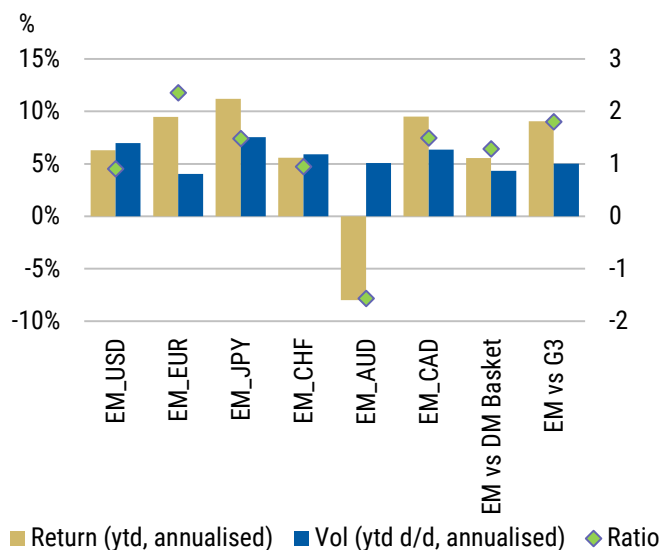
Source: Bloomberg, Morgan Stanley Research

Exhibit 23: Diversified funding reduces risk

Source: Bloomberg, Morgan Stanley Research

Exhibit 24: EM vs DM gains continue

Source: Bloomberg, Morgan Stanley Research

Exhibit 25: Long EM vs EUR has been effective, but vs G3 also attractive

Source: Bloomberg, Morgan Stanley Research

Which Currencies or Baskets Have Been the Most Effective Funders Over Longer-Run Trends?

While EUR-funded strategies have been the most successful so far this year, what about over a longer period?

We present some return statistics showing the pros and cons of funding EM currency exposure across:

- a range of different individual DM currencies (USD, EUR, JPY, CHF, AUD, CAD)
- three baskets of DM funding currencies, including: 1) an equally-weighted basket of these same 6 DM currencies; 2) an equally-weighted basket of the same currencies excluding USD, and 3) a basket of USD, EUR, and JPY - DM currencies that are more defensive in nature.

Our return statistics examined:

- the cumulative returns of all funding strategies, comparing the non-USD funding strategies vs USD funding
- Volatility, Sharpe ratios, max drawdown, Calmar, hit ratios, worst 3m performance, and the worst 12m performance.

The full sample of our return analysis runs from January 2010 to May 2026, and [Exhibit 26](#) shows the full return statistics across the various funding strategies. The results are just indicative and are for comparative purposes, without taking into account any transaction costs or strategies for implementation. The results should be self explanatory but for clarity:

- The Calmar ratio shows the annualised returns divided by the absolute value of the maximum drawdown - an indicator of overall return vs portfolio stress.
- The hit rate shows the number of positive return periods divided by the total number of periods - we are using daily data so a hit rate of 53% suggests returns were positive on 53% of trading days.

[Exhibit 27](#) shows specifically the Sharpe ratios of the different strategies for funding long EM exposure over different horizons. In addition to the full sample since 2010, we also calculated the return statistics for the last 1, 3, and 5 years. Given the variety of FX market conditions over these time periods, conclusions that are consistent across the different timeframes could be more robust.

Baskets outperform: Over the full sample since 2010, the funding EM exposure via baskets results generally produced more consistent outcomes than any single funding currency approach. Over the full sample, all 3 basket funding approaches generated significantly higher Sharpe ratios and materially lower drawdowns than the traditional USD-funded approach.

Importantly, this advantage persisted across the 5-year and 3-year horizons, suggesting that the benefit is not simply a consequence of the long USD bull market from 2010 to 2022. Not taking into account transaction costs, of course, artificially boosts the relative attractiveness of baskets, as these approaches would likely have involved higher costs.

Among the individual funding currency strategies, CAD stands out. In the full sample since 2010, long EM versus CAD delivered an annualized total return of 3.6%, volatility of 6.3%, and a Sharpe ratio of 0.58, which was close to the best basket result. It also had the lowest maximum drawdown among the individual funders at -12.3%, compared with -28.8% for USD, -25.2% for JPY, and -28.7% for CHF.

CAD also had the best worst-3-month and worst-12-month results among the single funders, at around -8.2% and -8.4%, respectively. This suggests that CAD funding historically provided a more balanced exposure: it improved return versus USD while not

introducing the same degree of left-tail risk as JPY or CHF.

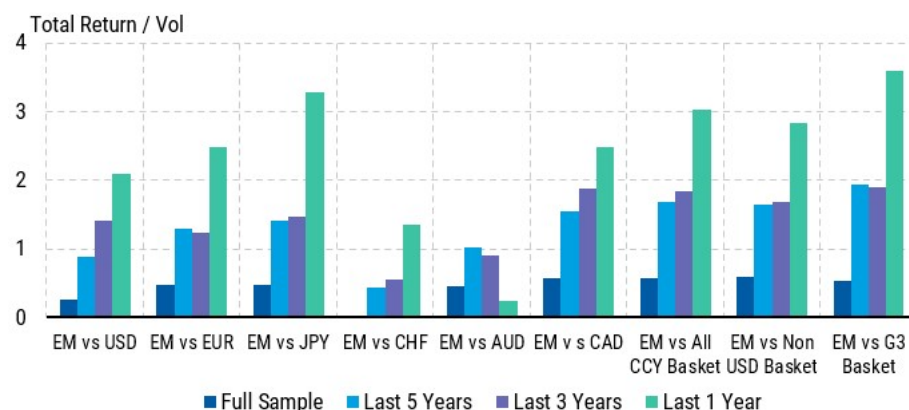
With the exception of the last 1y, funding long EM exposure in CAD has produced higher Sharpe ratios than the other individual funding currency choices across all timeframes.

Exhibit 26: Return statistics for differently funded long EM currency exposure (since 2010)

Strategy	Ann. Return	Ann. Vol	Sharpe	Max Drawdown	Worst 3M	Worst 12M	Calmar	Hit Rate	Total Return
EM_USD	1.9%	7.3%	0.26	-28.8%	-15.5%	-18.4%	0.07	52.4%	37.2%
EM_EUR	3.2%	6.6%	0.48	-15.5%	-12.8%	-12.3%	0.20	52.4%	69.7%
EM_JPY	5.2%	10.7%	0.49	-25.2%	-16.9%	-19.8%	0.21	53.9%	136.8%
EM_CHF	0.3%	9.2%	0.03	-28.7%	-19.5%	-24.0%	0.01	51.4%	4.6%
EM_AUD	3.4%	7.5%	0.45	-18.5%	-11.2%	-17.8%	0.18	51.4%	75.0%
EM_CAD	3.6%	6.3%	0.58	-12.3%	-8.2%	-8.4%	0.30	52.9%	83.6%
Basket_All Equal	3.1%	5.4%	0.57	-12.6%	-11.9%	-10.2%	0.25	53.4%	67.8%
Basket_NonUSD Equal	3.3%	5.6%	0.59	-12.3%	-11.6%	-10.7%	0.27	53.8%	73.6%
Basket_G3_USD_EUR_JPY	3.7%	6.8%	0.54	-13.6%	-13.5%	-11.2%	0.27	59.6%	80.9%

Source: Bloomberg, Morgan Stanley Research

Exhibit 27: Funding with either baskets or CAD provided higher Sharpe ratios & more consistent returns



Source: Bloomberg, Morgan Stanley Research Note: Full Sample - Jan 2010 - Jun 2026

Conclusion

In this publication we illustrate the strong long-term performance of our FX Multi-Factor Quantitative Investment Strategies. The performance is robust across business cycles and market conditions. While baseline implementations of FX systematic investing yield strong returns we advocate for an evolution to a broader set of signals. Adding FX Multi-Factor to a portfolio of QIS as well as a traditional 60/40 portfolio leads to an improvement of risk-adjusted returns and a reduction of drawdown risks. Our analysis highlights the relevance of systematic signals for fundamental investors to guard against confirmation bias, to time and size trades, and for the filtering of market views.

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